

Module 4

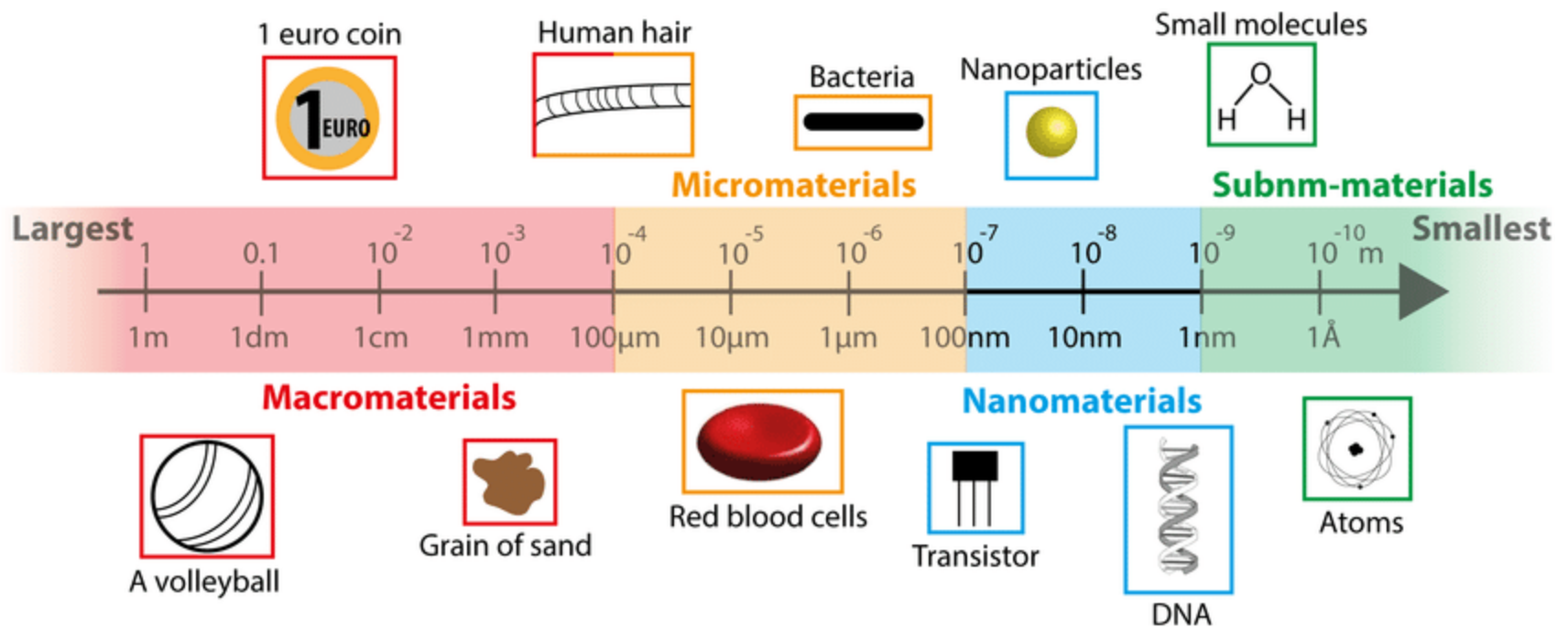
Engineering Materials

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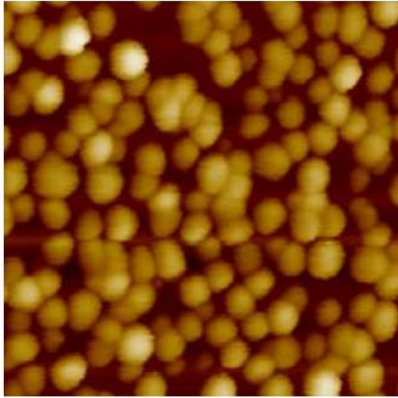
Nanomaterials

Nanotechnology: Designing Materials Atom by Atom

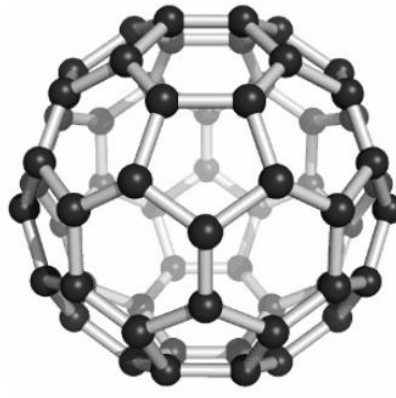
Nanotechnology is the science and engineering of nanoscale systems, whose critical sizes range from **1 to 100 nm**.



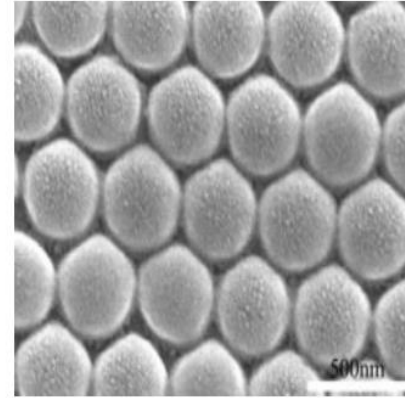
Examples of Nanomaterials



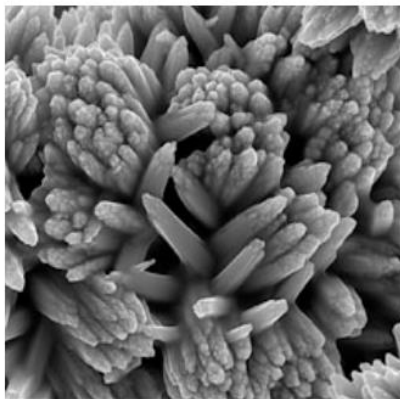
Au nanoparticle



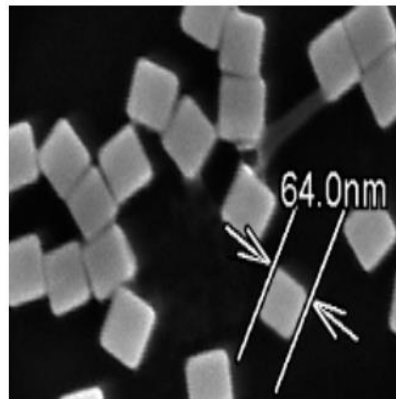
Buckminsterfullerene



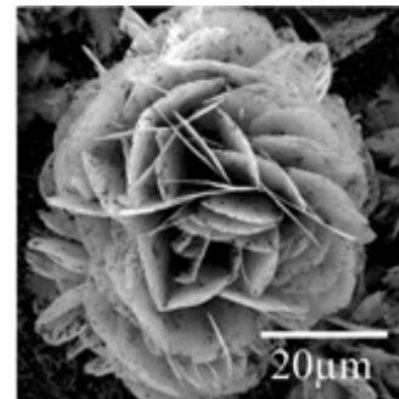
FePt nanosphere



Titanium nanoflower

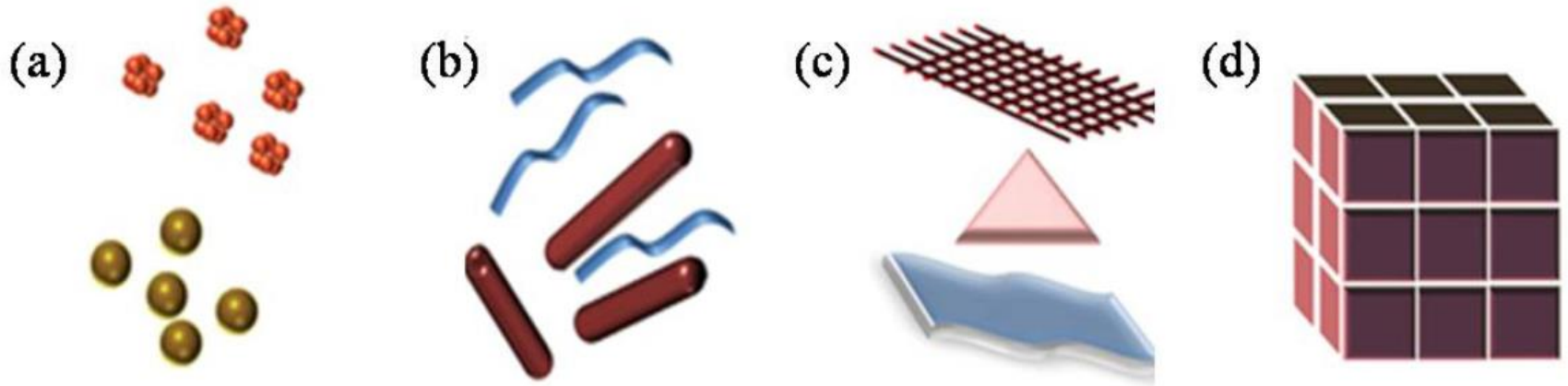


Silver nanocubes



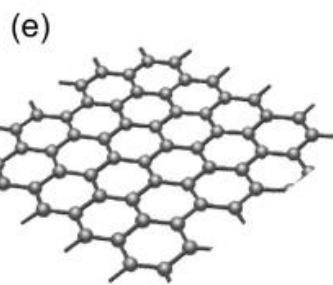
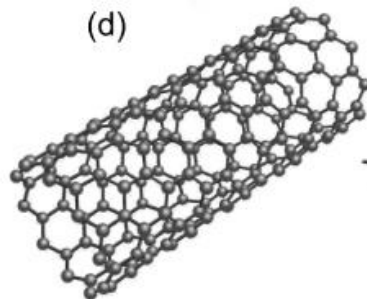
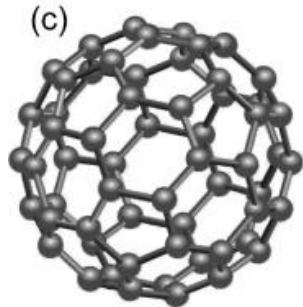
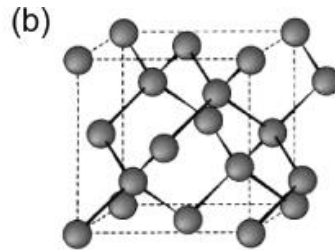
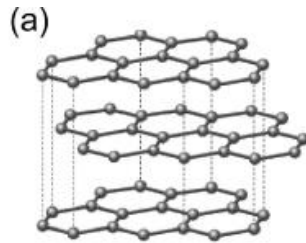
SnO₂ nanoflower

Examples of 0, 1, 2, 3-Dimensional Nanomaterials



Classification of Nanomaterials (a) 0D spheres and clusters, (b) 1D nanofibers, wires, and rods, (c) 2D films, plates, and networks, (d) 3D nanomaterials.

Examples of 0 to 3D Materials



Forms of carbon:

- a) Graphite 3D
- b) Diamond 3D
- c) Buckminsterfullerene
“Buckyballs” 0D
- d) Carbon nanotube 1D
- e) Graphene 2D

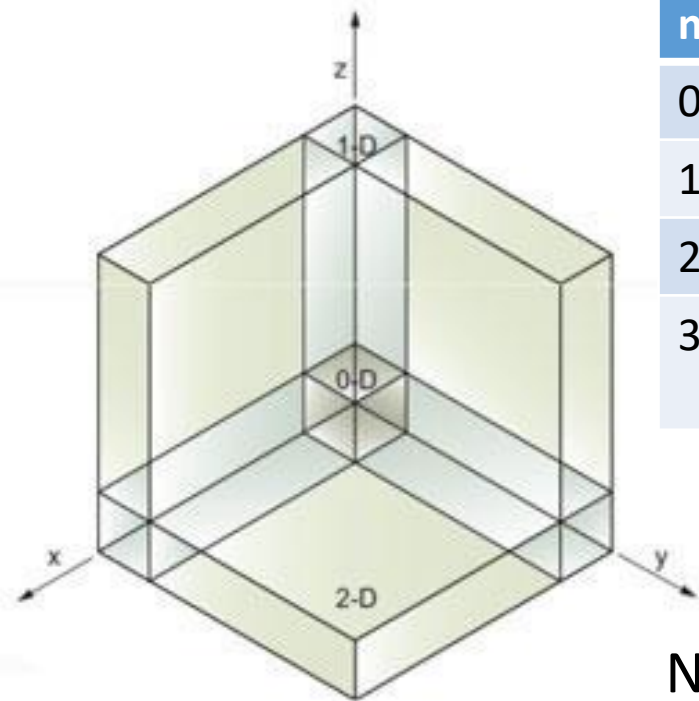
Bulk and nanoscale materials

The **key difference** between nanomaterials and bulk materials is that the **nanomaterials have their size in 1-100 nm** range at least in one dimension whereas the **bulk materials have their size above 100 nm** in all dimensions.

Classification (Macro scale)	Example	Nanoscopic Dimensions
Bulk (3D)	Anything you can see by eye	0
2D (nanosheet)	Graphene	1
1D (nanotube or nanowire)	Carbon nanotube	2
0D (nanoparticle)	Quantum dots (QDs)	3

Quantum Confinement Effect

Quantum confinement refers to the confinement of the electrons in nanoscale objects such as nanospheres, tubes, sheets, crystals.



Dimension at macroscale	Electron confinement at nanoscale	Example
0D	3D	CdSe-QD
1D	2D	Carbon nanotube
2D	1D	Graphene
3D	0D	Micron sized particle

Example

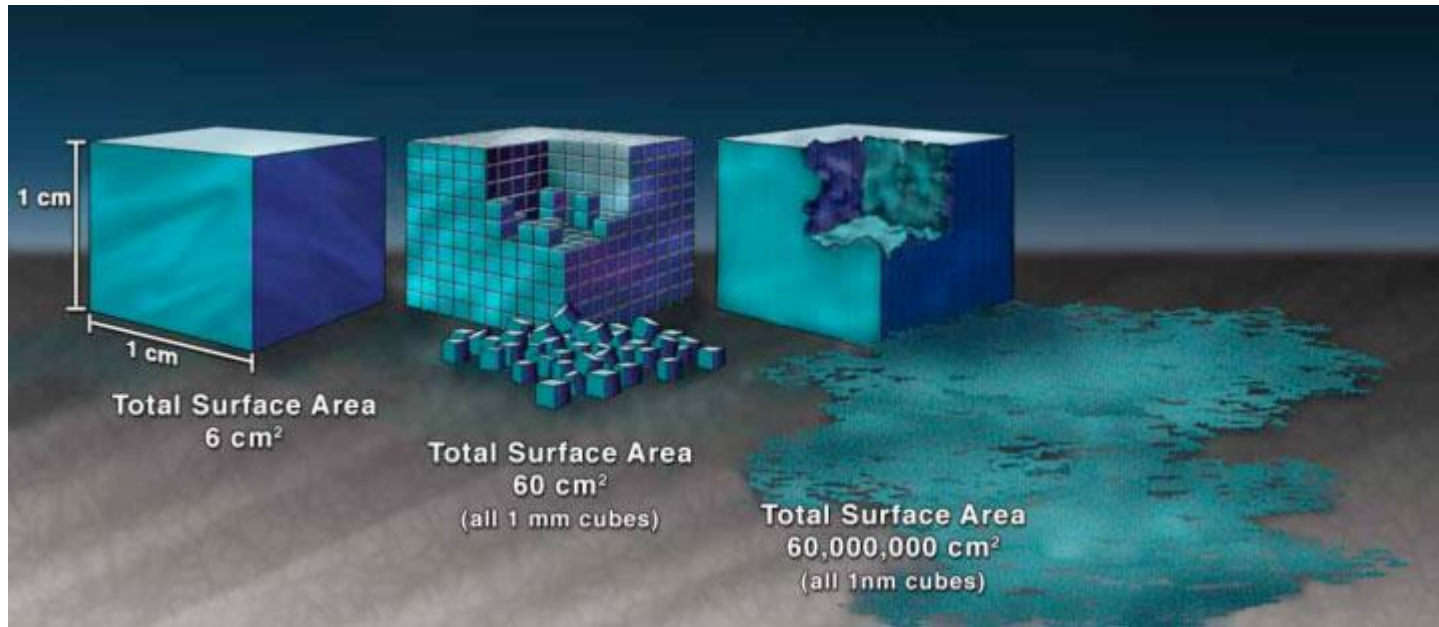
Nanosphere is a zero-dimensional material.
Therefore electrons are confined in 3-Dimension.

In what way the nanomaterials differ from bulk materials?

Two principal factors cause the properties of nanomaterials to differ significantly from other materials: *increased relative surface area*, and *quantum effects*. These factors can change or enhance properties such as reactivity, strength and electrical characteristics.

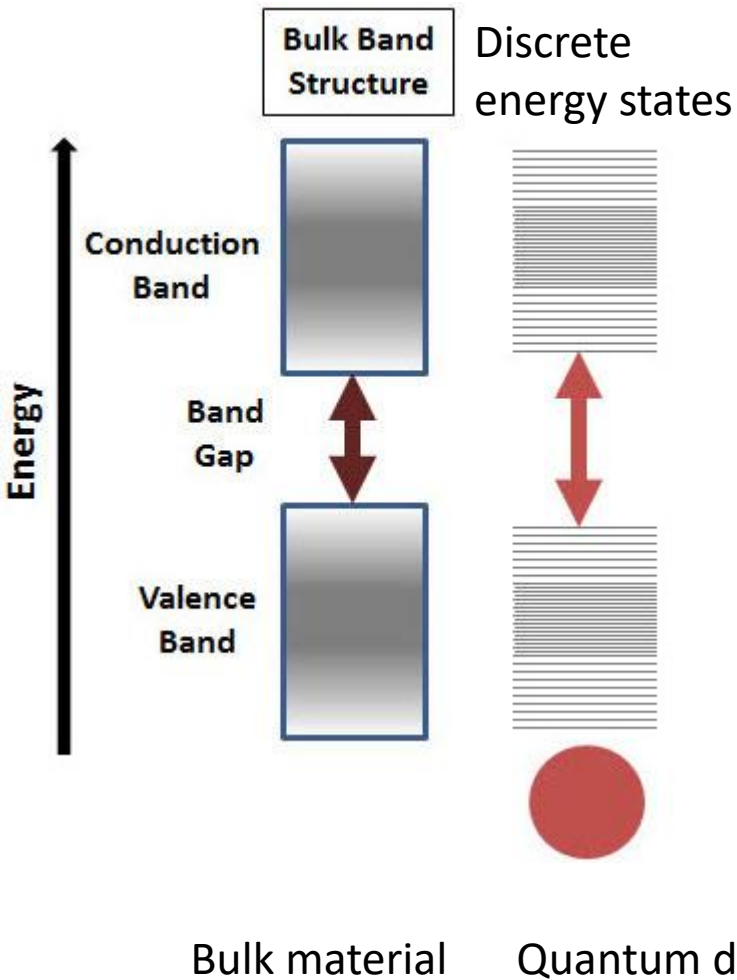
High-Relative Surface Area

Nanoscale materials have high surface area as compared to the bulk material.



As compared to the bulk materials, nanoscale catalysts can lead increased catalytic reaction rate, selectivity and catalytic efficiency of various chemical reactions.

Quantum confinements affects the energy levels and electrical conductivity



In bulk materials the electrons are not confined. They are free to move in all Directions to longer distance.

In quantum dots the electrons are confined in all dimension.

Due to electron confinement quantum dots have poor electrical conductivity as compared to bulk material

Quantum dots (QD).

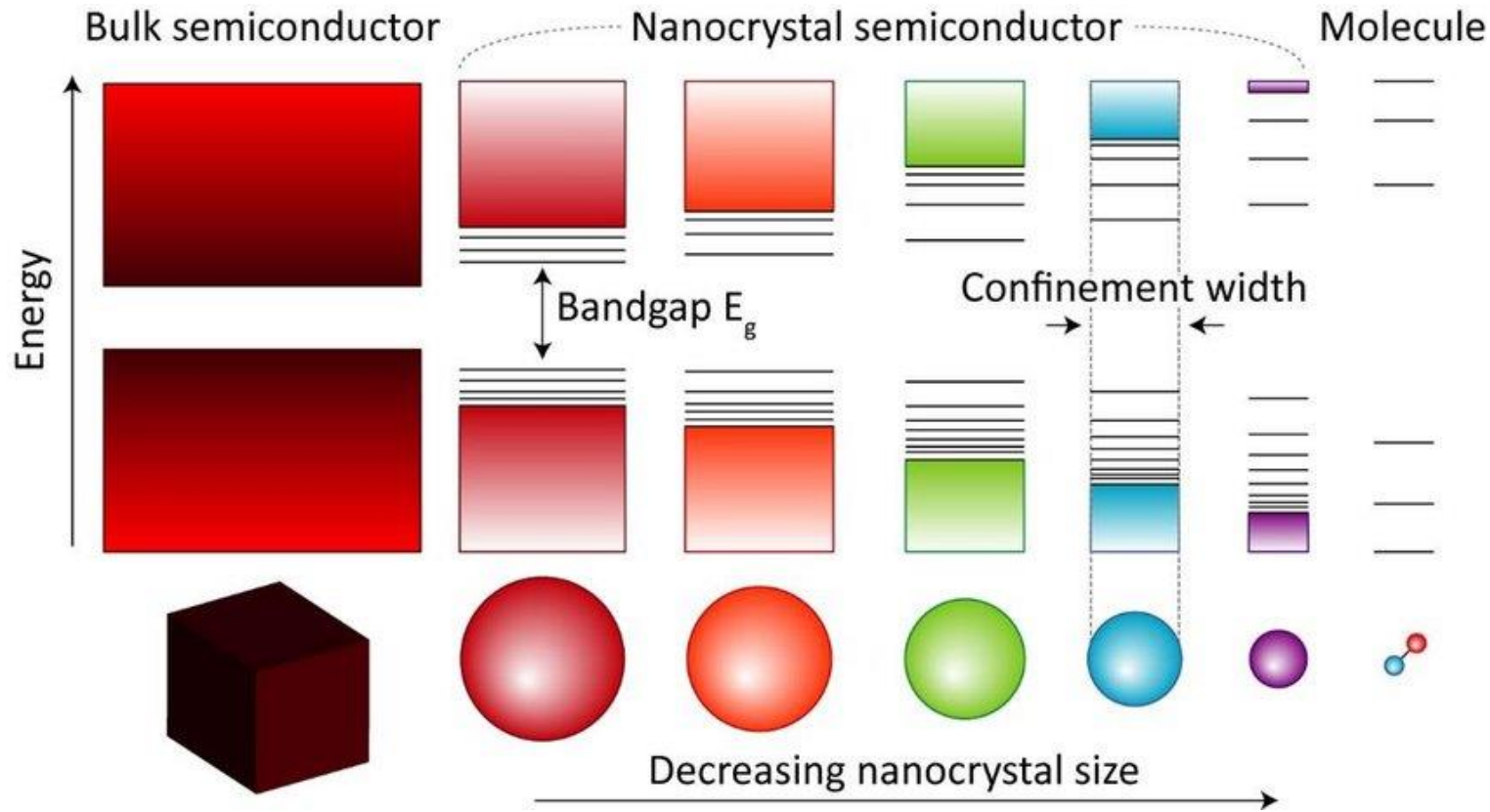
Quantum dots are semicrystalline nanoparticles.

Example: Cadmium Sulphide (CdS) and Cadmium Selenide (CdSe)

They show light absorbance and emission in the visible region of the electromagnetic spectrum.

The colour they absorb/emit depends on the size of the nanoparticle.

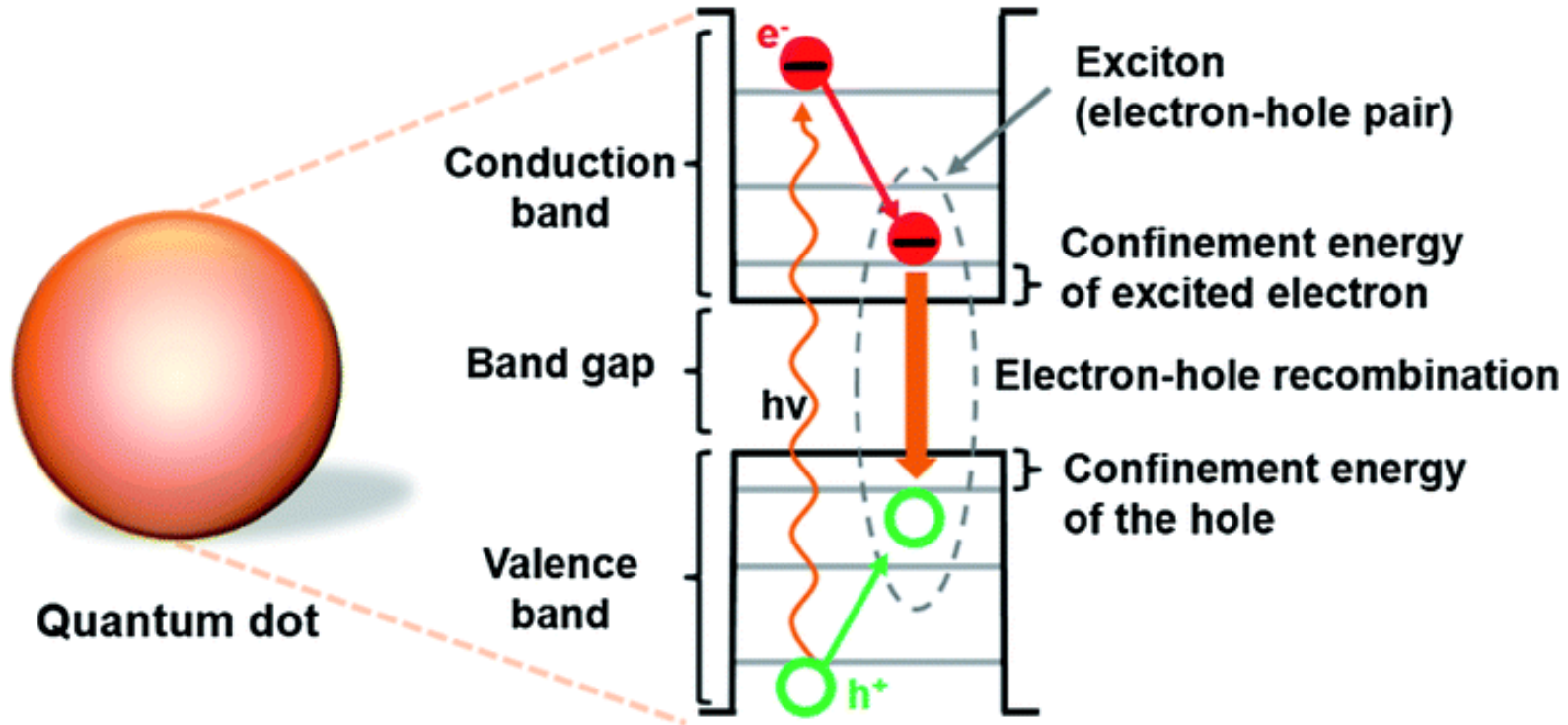
Relationship between the energy gap and size of the QD



As shown in figure, bulk semiconductors have energy bands. Whereas quantum dots have discrete quantized energy states. ***As size of the particle decreases the gap energy increases.***

Photoluminescence (PL) of quantum dots

Exciton, the combination of an electron and a positive hole (an empty electron state in a valence band), which is free to move through a nonmetallic crystal as a unit.



Exciton is formed by the absorbance of light. The electron-hole recombination is accompanied by the emission of light which is called **fluorescence**.

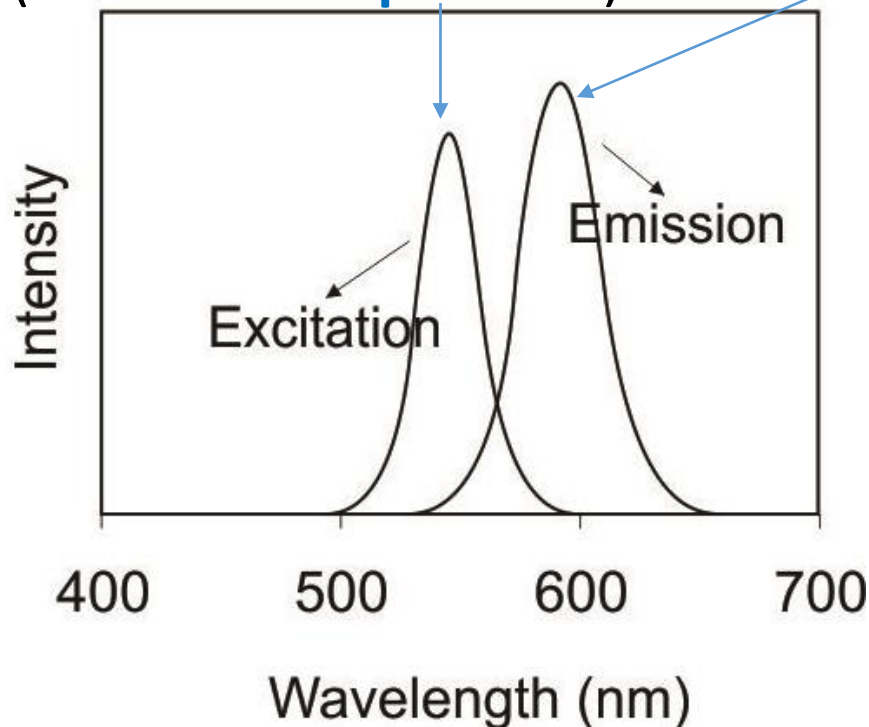
Absorbance and Emission spectrum

The absorption and emission light by material can be quantitatively determined and one can construct a spectrum (***intensity of light as a function of wavelength***).

Input light

(**absorbance spectrum**)

Output light (**emission spectrum**)



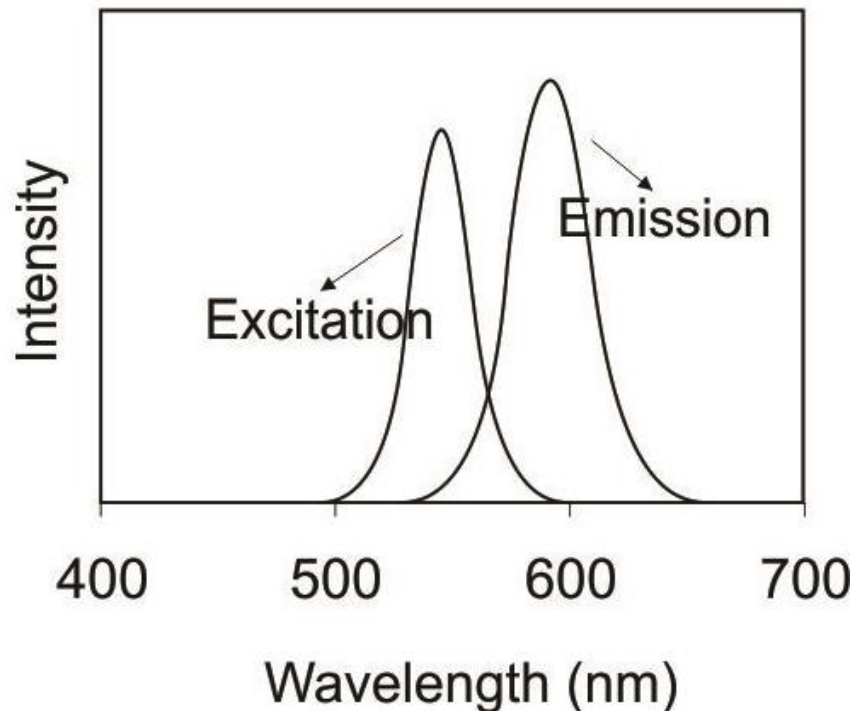
The process of light input to the material is called excitation.

The process of light output from material is called emission.

Absorbance and Emission spectrum

The maxima of emission spectrum (middle wavelength) is always larger than that of absorption maximum.

The emitted light will have less energy than the energy of the light being absorbed. ***It is due to the loss of small energy as heat.***



$$E = hc/\lambda$$

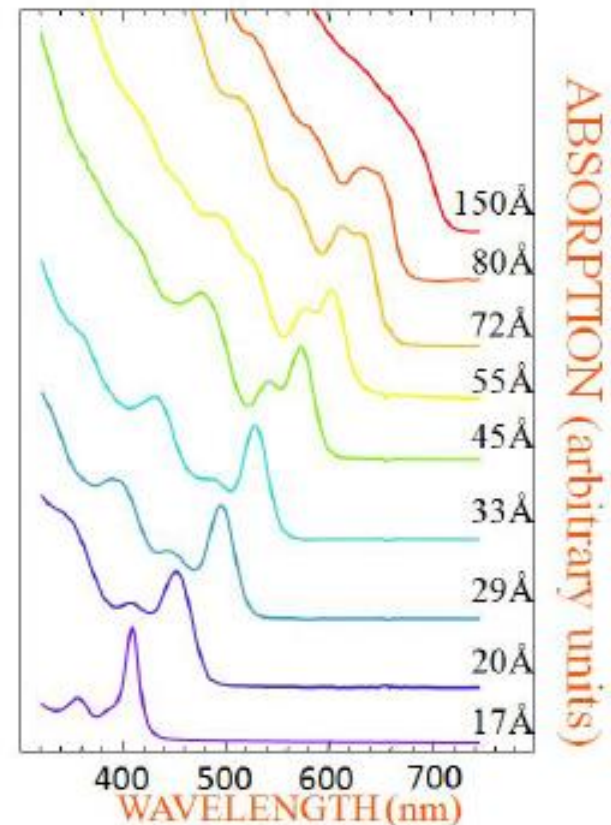
Longer wavelengths
– smaller energy

Smaller wavelengths
– larger energy

Color of emission and size of the quantum dots

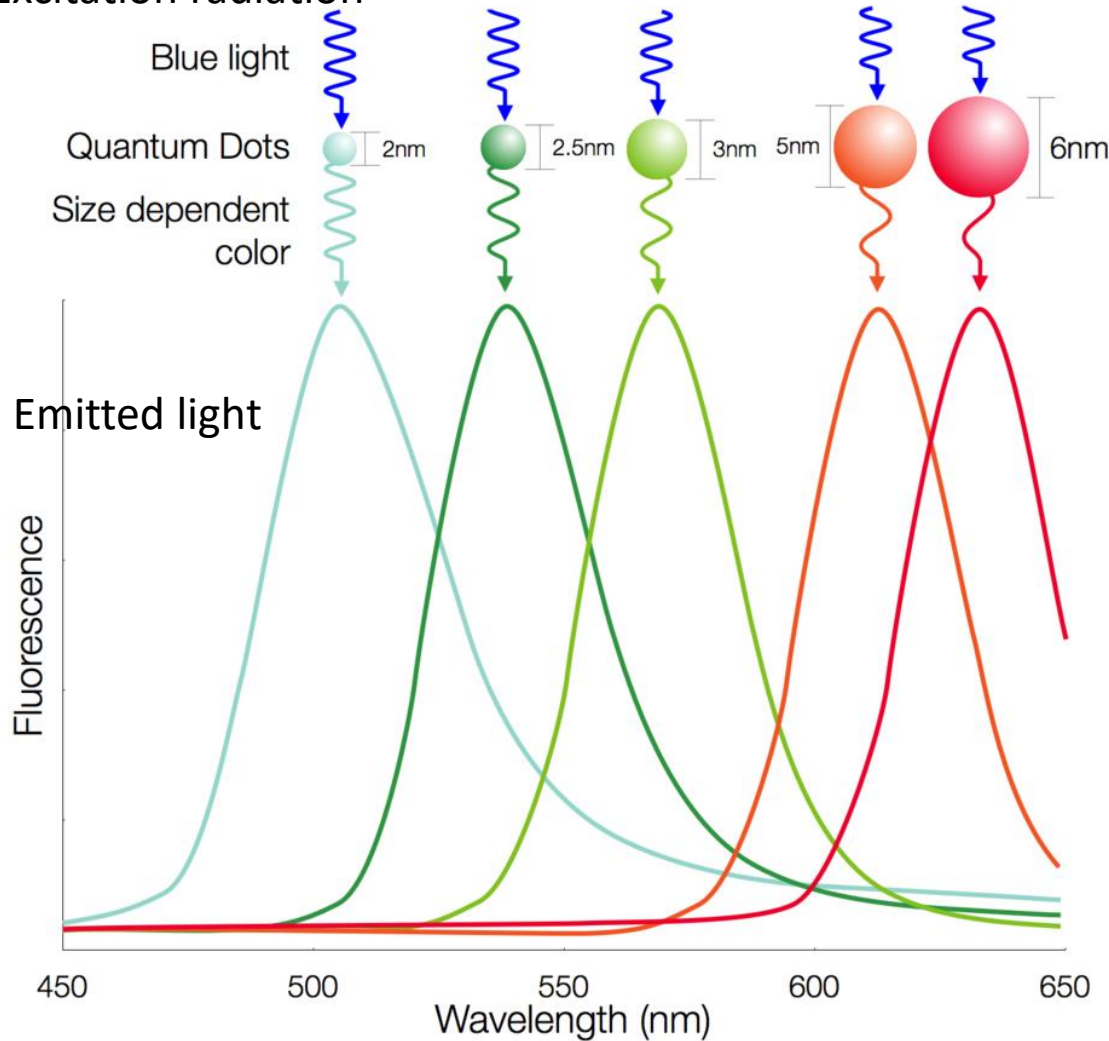
Quantum dots absorb and emit light.

Smaller particle absorbs and emits at *lower wavelength* of the visible radiation.



Size dependent emission spectrum of QD

Excitation radiation

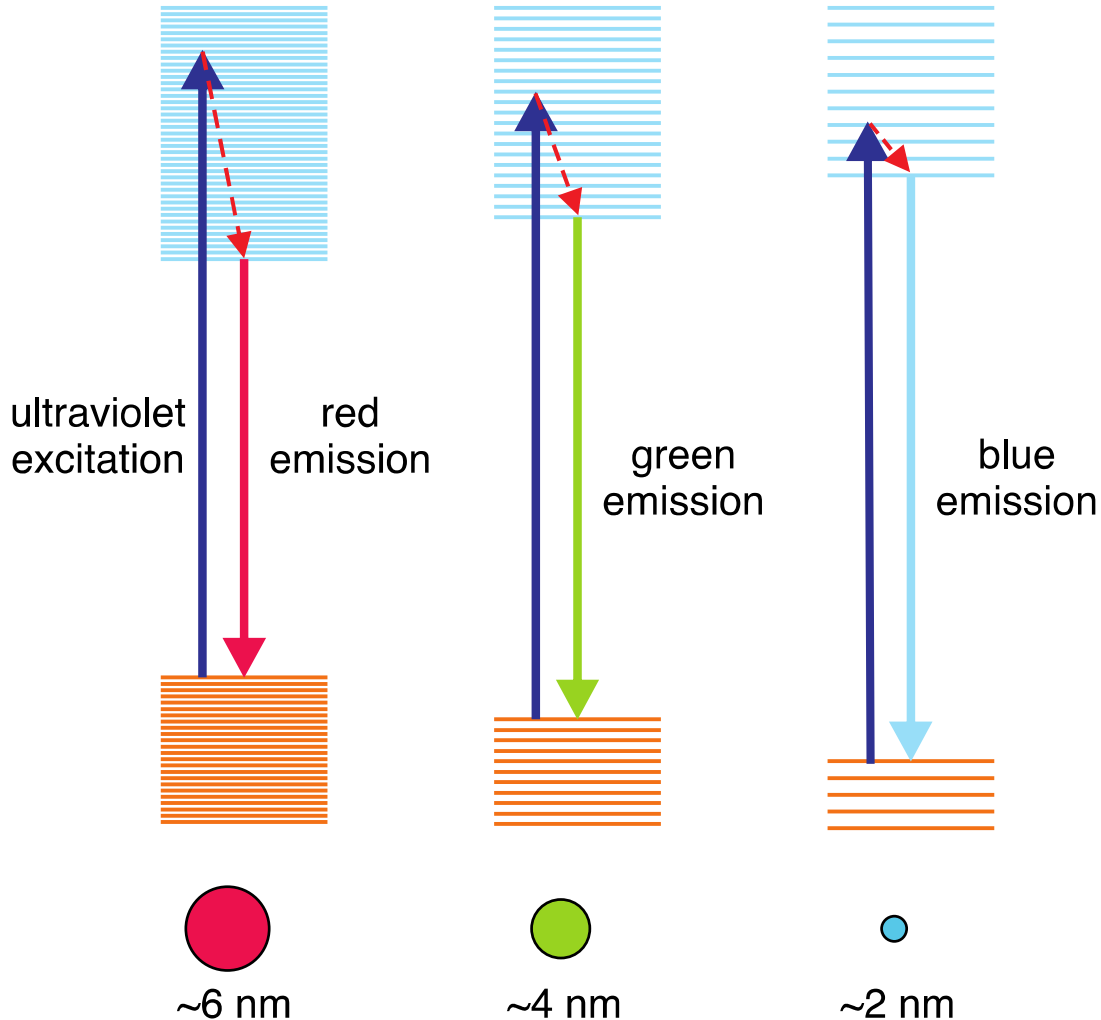


Energy of emitted light (fluorescence) < absorbed light,

So, one can use high energy blue light to excite any sized QD.

<https://physicsopenlab.org/2015/11/20/quantum-dots-a-true-particle-in-a-box-system/>

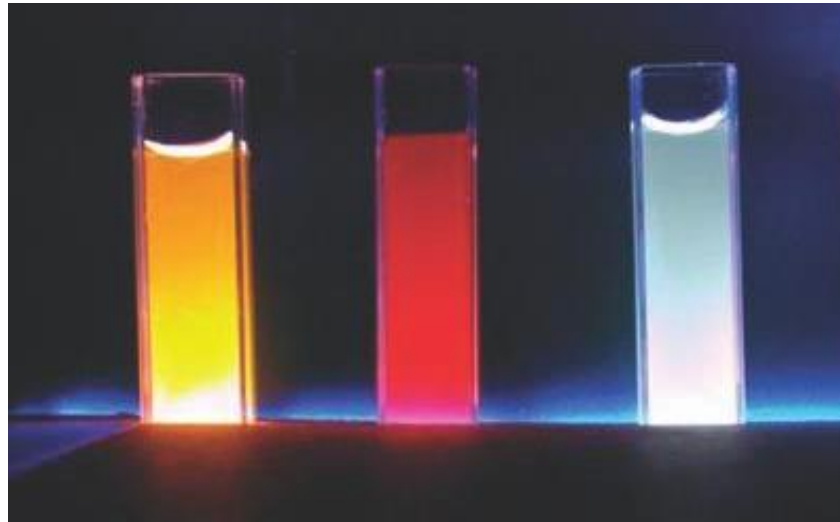
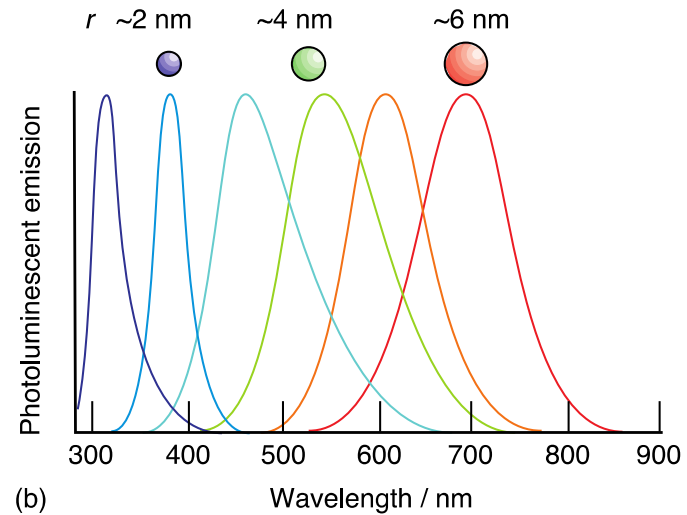
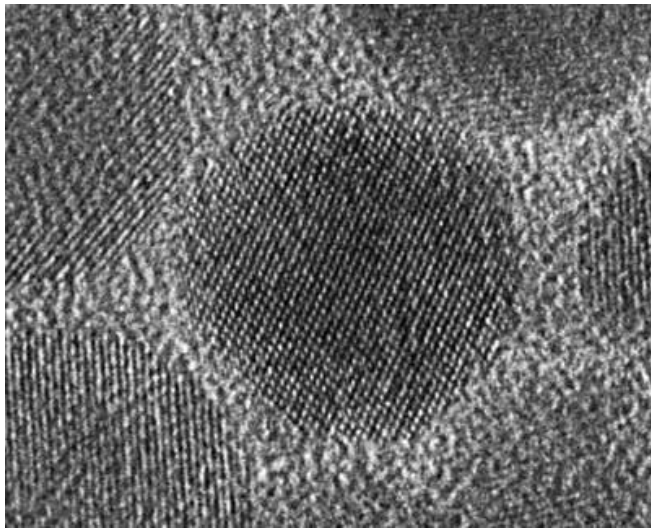
Same absorbance light-Varied emission light



Smaller particle,
larger the band gap.

small wavelength
light is emitted

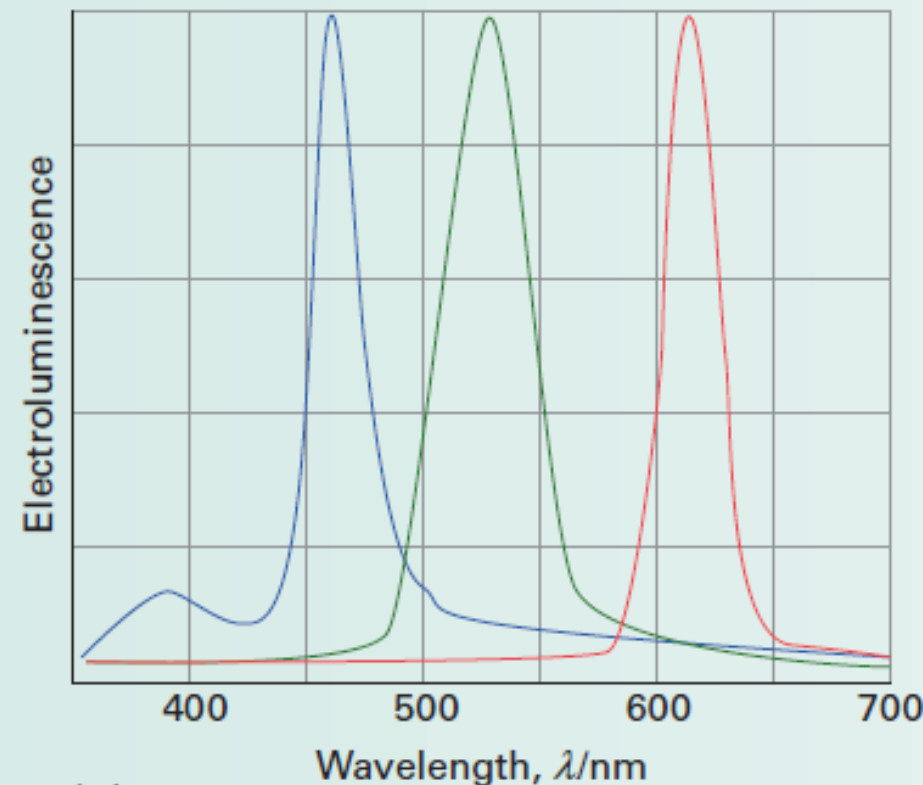
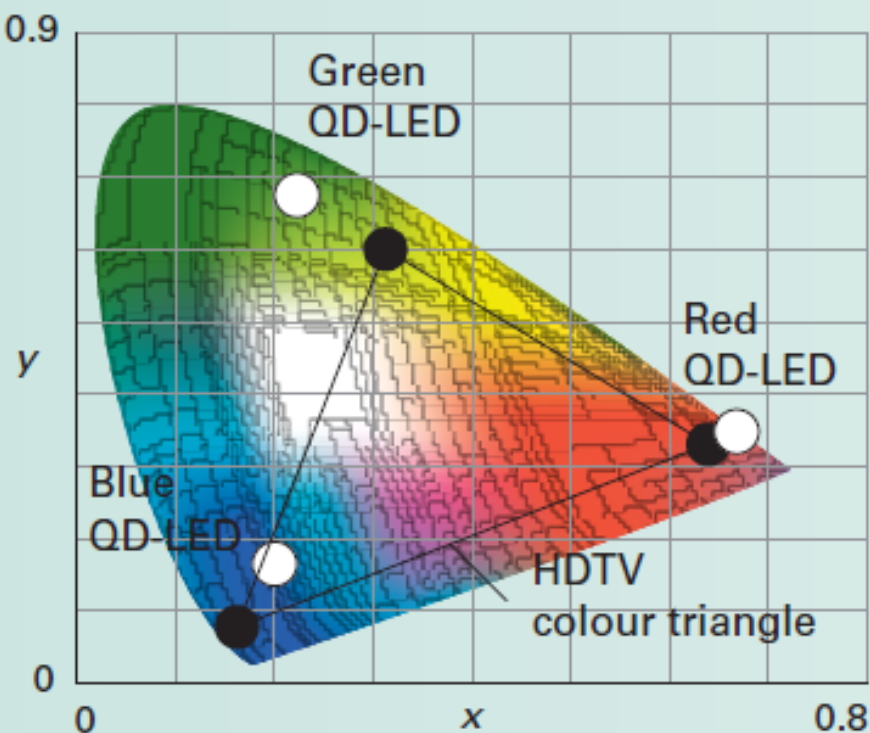
Larger energy is
absorbed or emitted



Cadmium sulfide quantum dots: (a) a CdS dot approximately 8nm diameter; (b) the photo- luminescent colours emitted by CdS quantum dots, schematic; (c) Photoluminescence of two different sized quantum dots (left and centre) and a mixture of the two, giving an approximately white output

Chromaticity diagram shows the positions of red, green, and blue QD-LED colours, an HDTV colour triangle is shown for comparison.

Normalized EL spectra of fabricated QD-LEDs corresponding to the colour coordinates



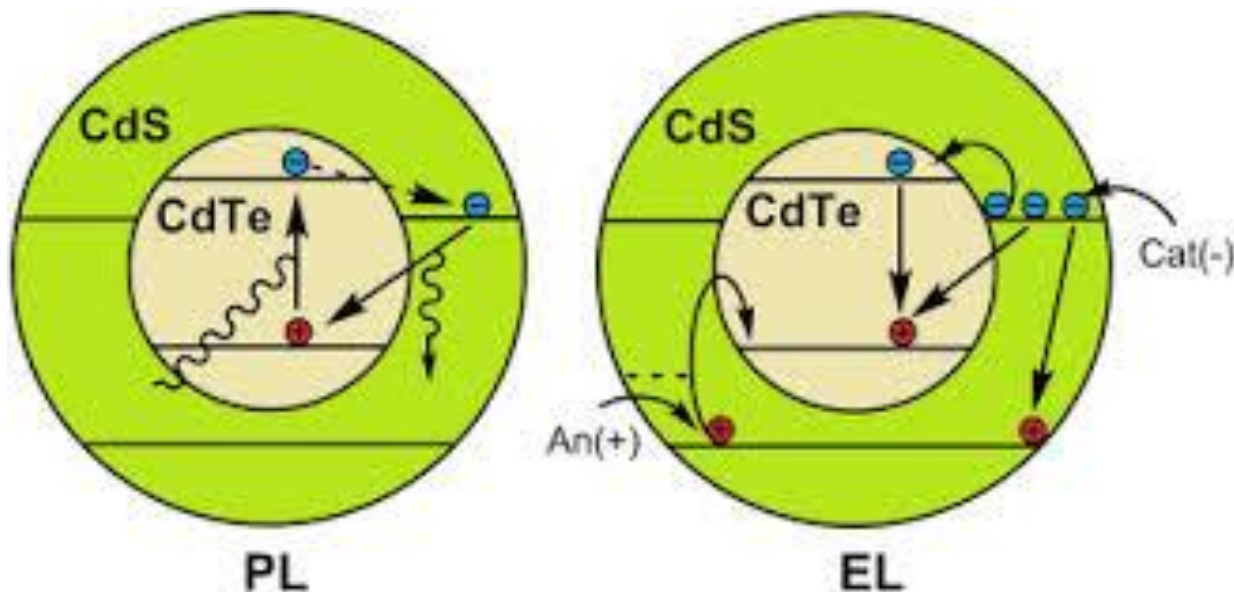


A typical flat-screen TV (left) can only express about 20% of the colors that are detectable with the human eye. A QD-enhanced TV (right) can produce over 60% of the spectrum seen by the human eye.

Reason:

The half-width of the emission spectrum produced by normal LED is wider than the half-width of the emission spectrum produced by the quantum dots (compare the inset figure showing visible region spectrum produced by LED and QD)

Photoluminescence (PL) Vs electroluminescence (EL) of quantum dots



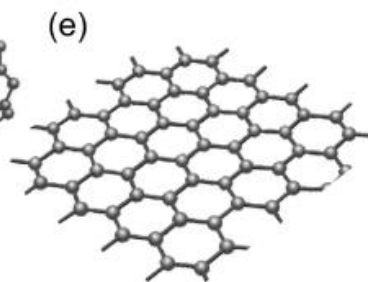
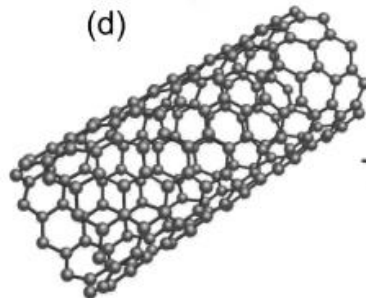
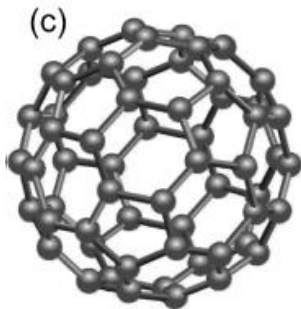
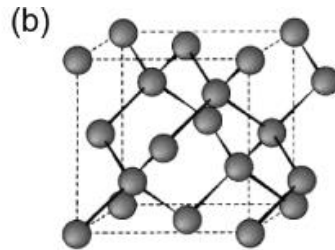
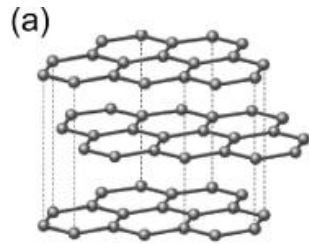
In photoluminescence the **excitons (electron-hole pair)** are produced by the **absorbance of light**.

In electroluminescence excitons are produced by the **application of potential bias**.

Applications of Nanoparticles

- Silver nanoparticles have good antibacterial properties, and are used in surgical instruments, refrigerators, air-conditioners, water purifiers etc.
- Gold nanoparticles are used in catalytic synthesis of silicon nano wires, sensors for the detection of tumors, drug delivery cargos.
- ZnO nanoparticles are used in electronics, ultraviolet (UV) light emitters, piezoelectric devices and chemical sensors.
- TiO_2 nanoparticles are used as photocatalyst and sunscreen cosmetics (UV blocking pigment).
- Antimony-Tin-Oxide (ATO), Indium-Tin-Oxide (ITO) nanoparticles are used in car windows, liquid crystal displays and in solar cell preparations.

Carbon Materials



Forms of carbon:

a) Graphite 3D

b) Diamond 3D

c) Buckminsterfullerene
“Buckyballs” 0D

d) Carbon nanotube 1D

e) Graphene 2D



2010 Nobel Prize in Physics



Andre Geim

http://www.nobelprize.org/nobel_prizes/physics/laureates/2010/

Konstantin Novoselov

Properties of graphene

Thinnest material possible (~0.345 nm thick)

Strongest material ever measured

- 200 times more stronger than steel

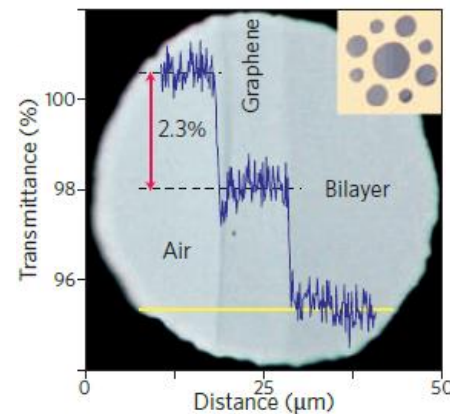
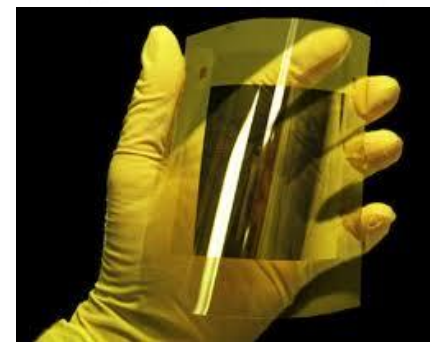
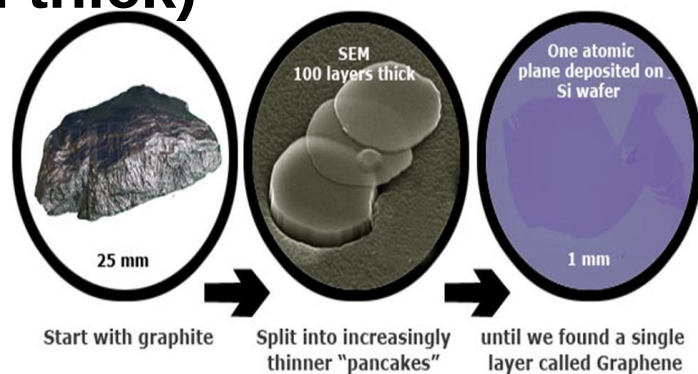
Electrical conductivity is best known

so far for any material

- 10,00,000 times more conductive than copper

Flexible and stretchable

Transparent (~97.7% white light can pass through a single layer of graphene)



Potential applications of graphene

Optoelectronics

LEDs, Touch screens,
Flexible displays etc.

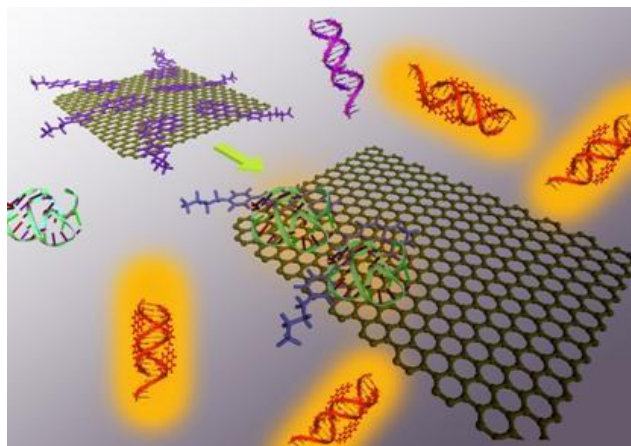


Energy generation and storage

Photovoltaics,
Supercapacitor etc.

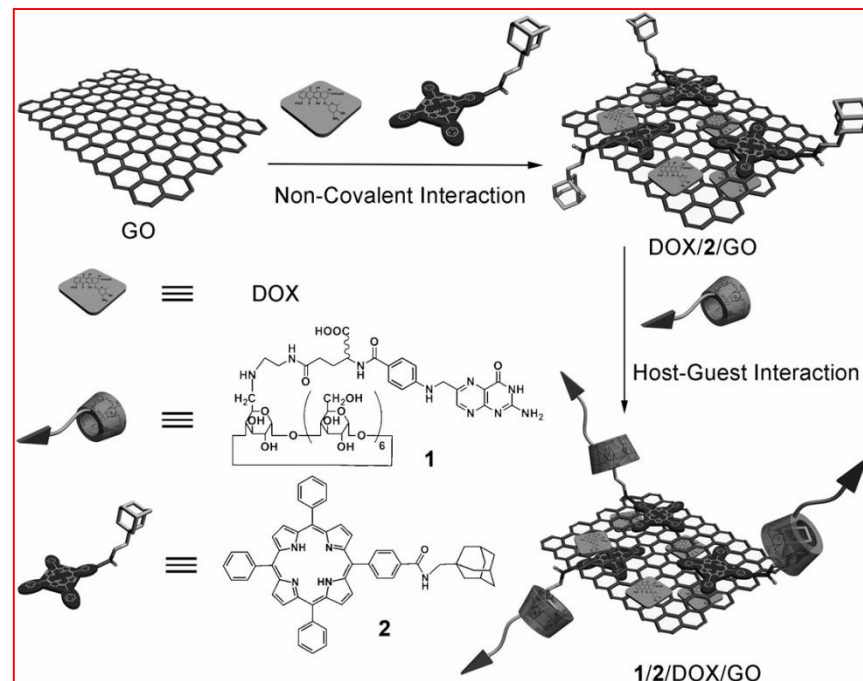
Drug delivery

Anticancer
drug: Doxorubicin



Sensors

DNA sensor



Chem. Eur. J. 2012, 18, 4208

Angew. Chem. Int. Ed. 2010, 49, 6549

Some questions

Energy conversion

1. The wave length of the emitted radiation is 468 nm, calculate the band gap energy of the material in kJ/mol

Solution 1

$$E_{\text{photon}} = \frac{h \cdot c}{\lambda}$$

The number of particles in one mole is given by the Avogadro constant, which is $N_A = 6.02 \cdot 10^{23} \text{ mol}^{-1}$. In this case the particles are photons.

As we have the expression for energy per photon, we need to multiply it by the Avogadro constant to get the amount of energy per one mole of photons.

$$E_{\text{per mole}} = \frac{N_A \cdot h \cdot c}{\lambda}$$

Note how the units for this expression cancel out to what we desire:

$$\frac{J \cdot s \cdot m}{s \cdot m \cdot \text{mol}} = \frac{J}{\text{mol}}$$

Using SI units (m , $\frac{m}{s}$, $J \cdot s$, mol^{-1}), substitute all the values into the equation to get the answer:

$$\begin{aligned} E_{\text{per mole}} &= \frac{6.02 \cdot 10^{23} \cdot 6.63 \cdot 10^{-34} \cdot 3.00 \cdot 10^8}{468 \cdot 10^{-9}} \\ &= 256 \cdot 10^3 \frac{\text{J}}{\text{mol}} \end{aligned}$$

Finally, convert joules to kilojoules:

$$E = 256 \frac{\text{kJ}}{\text{mol}}$$

Example 2. Convert 2eV to nm

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$E = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{E}$$

The wavelength of a 2 eV photon is given by:

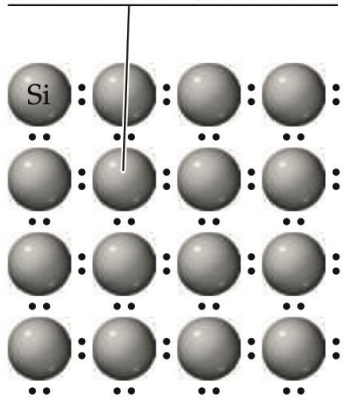
$$\lambda = hc / E_{ph} = 6.625 \times 10^{-34} \times 3 \times 10^8 / (1.6 \times 10^{-19} \times 2) = 621 \text{ nm.}$$

3. Identify each of the following as a p-type or n-type semiconductor:

- (a) aluminum doped silicon,
- (b) antimony doped germanium,
- (c) boron doped silicon,
- (d) Gallium doped germanium
- (e) Boron doped diamond

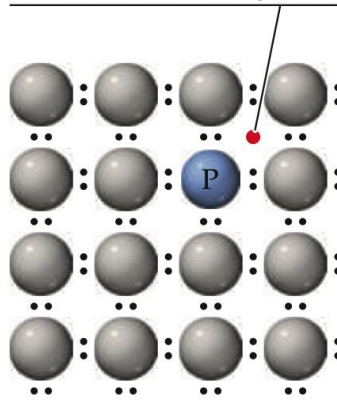
4. Illustrate the band structures of i, n, and p-type semiconductors

Silicon forms bonds to four other silicon atoms and has a complete octet.



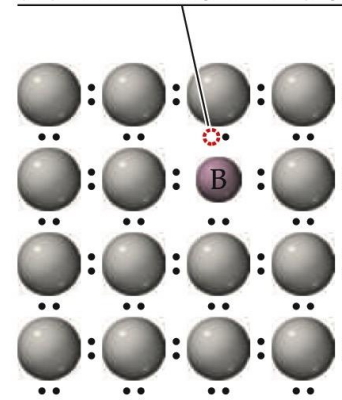
Si crystal

A silicon crystal doped with a P atom has an extra electron that is not needed for bonding. The *negative* electron is the charge carrier.



n-type semiconductor

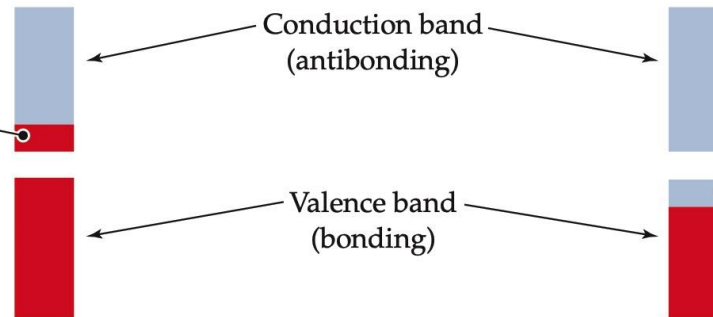
A silicon crystal doped with a B atom is electron deficient. The charge carrier is called a *positive* hole (represented by an empty circle.)



p-type semiconductor



An *n*-type semiconductor, such as silicon doped with phosphorus, has more electrons than needed for bonding and thus has more electrons in the partially filled conduction band than pure silicon.



A *p*-type semiconductor, such as silicon doped with boron, has fewer electrons than needed for bonding and thus has more vacancies—positive holes—in the valence band than pure silicon.

5) If Ge crystals has been doped with arsenic, it will behave as n-or p-type semiconductor?

Demonstrate the change with band theory with the help of drawing band structures.

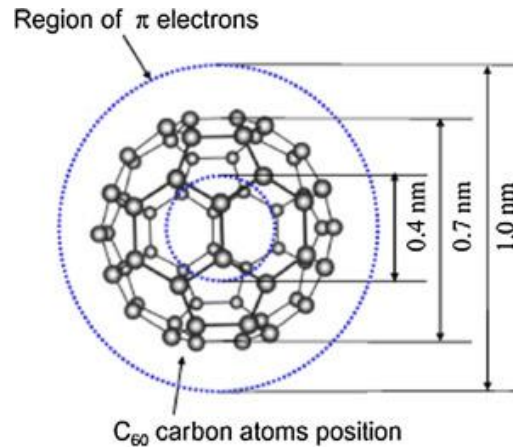
6) What is Exciton and how it is formed?

7) Why conductivity of semiconductor increases with increase in temperature?

8) Why conductivity of metal decreases with increase in temperature?

9) What determines the colour of the emitted light by the quantum dots?

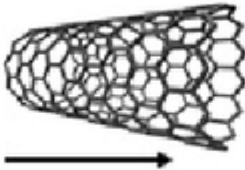
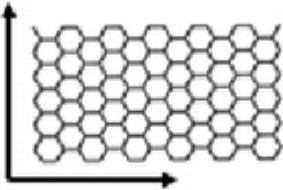
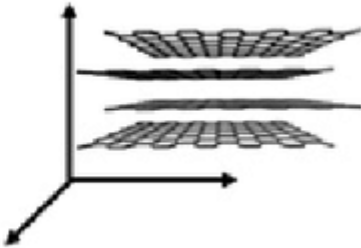
What is the macroscopic dimension of Buckminster fullerene (C_{60})?



Fullerene is a carbon allotrope. It is made up of 60 carbon atoms. All carbon atoms and its π -electron cloud is confined within 1 nm. It does not expand to micro or even bigger macro length scale. So, ***it is a 0D material at macroscopic scale*** (its electron cloud is absent beyond 100 nm in all dimension)

However, its dimension (**or electron confinement**) is 3D at nanometer scale (because its electron cloud is present in all directions within 100 nm from the center of the ball)

Classify fullerene, carbon nanotubes, graphene and graphite according to their macroscopic dimension?

0D	1D	2D	3D
			
Fullerene	Carbon Nanotube	Graphene	Graphite

All dimension the
electron is confined

In no direction the
electron is confined